MedTimeline — your family's health story, organized

DSH Hacks V2 · Theme: AI × Healthcare
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Purpose

When you care for an aging parent or manage a chronic condition, medical documents arrive scattered: a lab report from March, a follow-up in September, a discharge summary from a hospital visit. No one shows you the **story** — that HbA1c is trending up, that ferritin is falling, that a value flagged HIGH in September was borderline in March. Caregivers become accidental data managers, with no tools built for them. **MedTimeline turns scattered reports into one chronological health story** — with trends, flagged values, and questions for the next appointment — and answers every question with citations to the exact line.

Key features

- **Multi-document ingestion** — upload reports from any dates (PDF / TXT / MD); every passage keeps its exact origin.
- **Chronological timeline** — reports and measurements assembled into one dated story across all documents.
- **Trend detection** — repeated measurements compared across dates (e.g. "Ferritin: 22 ng/mL (2026-03) → 9 ng/mL (2026-09) — falling"), both values and dates shown.
- **Flagged values** — LOW/HIGH results surfaced with their reference ranges, straight from the documents.
- **Appointment prep** — auto-generated questions for the next visit, built from the trends and flags ("HbA1c rose from 5.6 to 5.9 — what could explain this?").
- 🔊 **Spoken questions** — one button reads the appointment questions aloud.
- **Grounded Q&A** — ask anything about the documents; answers cite the exact line and compare values across reports; off-document questions are declined.

Safety by design: MedTimeline is an organizer and translator, not a doctor. It never diagnoses, never prescribes, and does not interpret symptoms. Answers are restricted to retrieved excerpts of the uploaded documents; off-document questions return an explicit refusal. Appointment questions are phrased as questions *for your clinician*. The bundled sample reports are entirely synthetic (fictional patient data), and documents stay on the user's machine in this prototype.

How it works

Python + FastAPI backend; single-page web UI (no build step). Timeline extraction parses dated report rows into events, groups repeated measurements, and computes trends (rising/falling/stable) with source links. Retrieval is scikit-learn TF-IDF (local, any language); generation is retrieval-scoped with mandatory citations and an explicit refusal token. Audio uses edge-tts with content-hash caching. Any OpenAI-compatible LLM endpoint can be configured.

Impact

- **Caregivers** — the person managing a parent's care finally sees the whole story.
- **Continuity** — trends across providers and dates, which no single report shows.
- **Preparedness** — arrive at appointments with specific, document-grounded questions.
- **Safety** — grounded answers, honest refusals, and clinician handoff.

python fastapi scikit-learn edge-tts LLM RAG caregiving

Demo video: youtu.be/OVDilJSLxIQ (unlisted) · Code: github.com/maebahesioru/medtimeline
MedTimeline is not a medical device and is not for emergencies. Always consult a qualified clinician.